

② $a^2 - b^2 \rightarrow n(\text{odd})$

$$n = 2k+1 = \underbrace{k}_{\downarrow b} + \underbrace{(k+1)}_{\downarrow a}$$

$$\begin{aligned} a^2 - b^2 &= (a+b)(a-b) \\ &= ((k+1)+k)((k+1)-k) \\ &= (2k+1)(k-k+1) \\ &= (2k+1) \cdot 1 = 2k+1 = n(\text{odd}) \end{aligned} \quad \text{or}$$

$$\begin{aligned} a^2 - b^2 &= (k+1)^2 - k^2 \\ &= (k^2 + 2k + 1) - k^2 \\ &= \cancel{k^2} - \cancel{k^2} + 2k + 1 \\ &= 2k+1 = n(\text{odd}) \end{aligned}$$

③ $n + k \rightarrow m$
(r) (irr) (irr)

We assume, $n + k \rightarrow m \Rightarrow \begin{matrix} k \rightarrow m-n \\ \text{(irr)} & \text{(r)} & \text{(r)} \end{matrix}$

$$m = \frac{a}{b \neq 0} \quad n = \frac{c}{d \neq 0}, \quad \frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd} = \frac{4}{v}$$

$(m-n) + n \rightarrow \textcircled{m} \rightarrow \text{rational}$

④ 1) proof $\sqrt{2}$ is irrational

$\frac{a}{b \neq 0}$ (irreducible)

$\sqrt{2}$ is irrational

$$\left(\frac{a}{b}\right)^2 = (\sqrt{2})^2$$

$\frac{\text{even}}{\text{even}}$ (reducible)

$$\frac{a^2}{b^2} = 2$$

$$2b^2 = a^2 \quad (a \text{ is even})$$

$$2b^2 = (2k)^2 \Rightarrow 2b^2 = 4k^2 \Rightarrow b^2 = 2k^2 \Rightarrow b(\text{even})$$

2) disprove $\text{irr} * \text{irr} \rightarrow \text{irr}$

$$\sqrt{2} * \sqrt{2} = \underline{2} \text{ (rational)}$$

$$(5) \quad x + y \geq 2 \rightarrow x \geq 1 \text{ or } y \geq 1$$

$$\neg(x \geq 1 \text{ or } y \geq 1) \rightarrow \neg(x + y \geq 2)$$

$$x < 1 \text{ and } y < 1 \rightarrow \underline{x + y < 2}$$

$$(6) \text{ a) if } \underline{n^3 + 5 \text{ (odd)}} \rightarrow \underline{n \text{ (even)}}$$

by contraposition,

$$\underline{n \text{ (odd)}} \rightarrow \underline{n^3 + 5 \text{ (even)}}$$

$$\begin{aligned} 2k+1 &\rightarrow (2k+1)^3 + 5 \\ &(2k+1)(2k+1)(2k+1) + 5 \\ &(8k^3 + 12k^2 + 6k + 1) + 5 \\ &\underline{2(4k^3 + 6k^2 + 3k + 3)} \\ &2r \text{ (even)} \end{aligned}$$

b)

by contradiction,

$$\begin{aligned} &\neg(n^3 + 5 \rightarrow n) \\ &\neg(\neg(n^3 + 5) \vee n) \end{aligned}$$

$$\begin{aligned} &\frac{n^3 + 5 \wedge \neg n}{\text{(odd)} \quad \text{(odd)}} \\ &\frac{(2k+1)^3 + 5}{2(4k^3 + 6k^2 + 3k + 3)} \quad 2k+1 \\ &2r \text{ (even)} \neq \text{odd} \end{aligned}$$

$$(7) \quad \forall x \exists M, \quad \underline{S(B, x)} \leftrightarrow \neg S(x, x)$$

barber shaves a man iff man doesn't shave himself

$$\underline{S(B, B)} \leftrightarrow \neg S(B, B)$$

barber shaves himself iff doesn't shave himself

$$\Rightarrow$$

$P \leftrightarrow \neg P$		
	P	$\neg P$
T	F	F
F	T	F

8) 1) $x(\text{odd}), y(\text{odd})$

2) $x(\text{odd}), y(\text{even}) \equiv x(\text{even}), y(\text{odd})$

3) $x(\text{even}), y(\text{even})$

1) $xy \Rightarrow (2k+1)(2k+1) = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2r + 1 (\text{odd})$

$x+y \Rightarrow (2k+1) + (2k+1) = 4k + 2 = 2(2k+1) \text{ even}$

2) $xy \Rightarrow (2k+1)(2k) = 4k^2 + 2k = 2(2k^2 + k) = 2r \text{ even}$

$x+y \Rightarrow (2k+1) + (2k) = 4k + 1 = 2(2k) + 1 = 2r + 1 \text{ odd}$

3) $xy \Rightarrow (2k)(2k) = 4k = 2(2k) = 2r \text{ even}$

$x+y \Rightarrow (2k) + (2k) = 4k = 2(2k) = 2r \text{ even}$

True